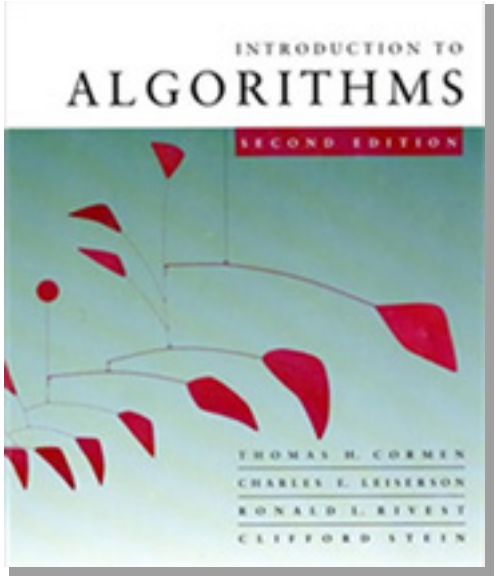
# *Introduction to Algorithms*

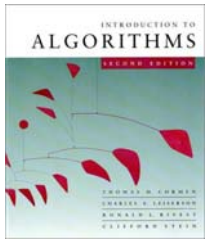



## LECTURE 14

**Amortized Analysis**

- Dynamic tables
- Aggregate method
- Accounting method
- Potential method


*Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson*



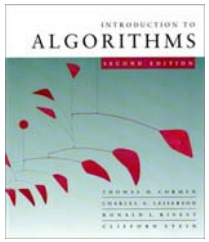

# Idea of amortized analysis

**Idea:** Average cost over a sequence of operations, because the worst case can not occur in each operation.

**Probabilistic analysis:**
average over all possible inputs for one operation. If using probability, called expected running time.

**Amortized analysis:**
sequence of operations, even expensive operation . No involvement of probability.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 




# How large should a hash table be?

**Goal:** Make the table as small as possible, but large enough so that it won't overflow (or otherwise become inefficient).

**Problem:** What if we don't know the proper size in advance?

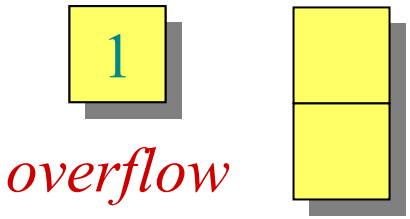
**Solution:** ***Dynamic tables.***

**IDEA:** Whenever the table overflows, "grow" it by allocating (via **`malloc`** or **`new`**) a new, larger table. Move all items from the old table into the new one, and free the storage for the old table.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Example of a dynamic table

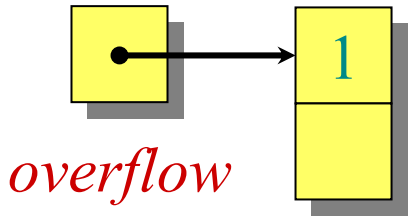
1. INSERT
2. INSERT




 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Example of a dynamic table

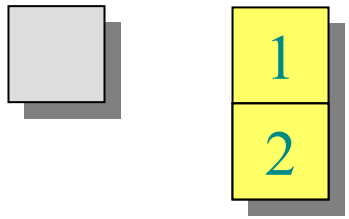
1. INSERT
2. INSERT




 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Example of a dynamic table

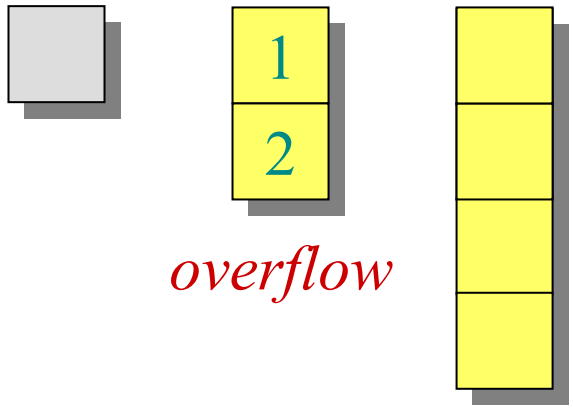
1. INSERT
2. INSERT




 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Example of a dynamic table

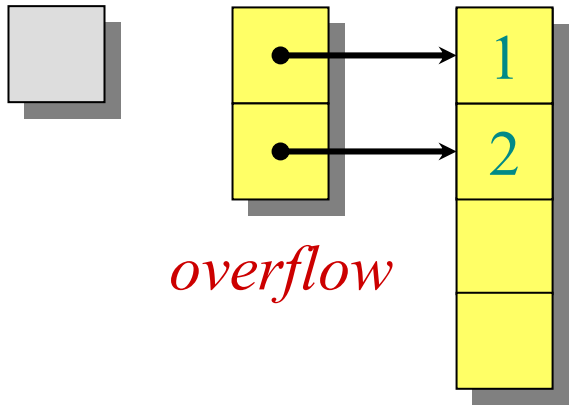
1. INSERT
2. INSERT
3. INSERT




 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Example of a dynamic table

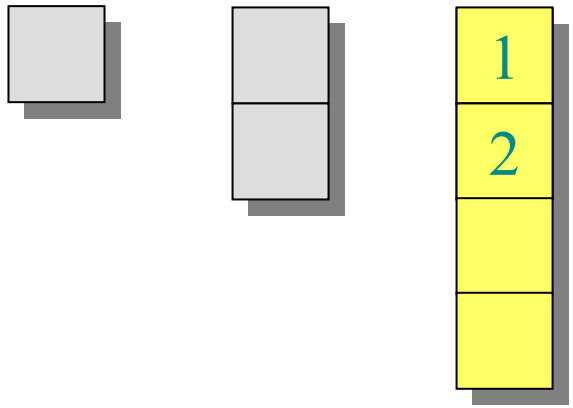
1. INSERT
2. INSERT
3. INSERT




 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Example of a dynamic table

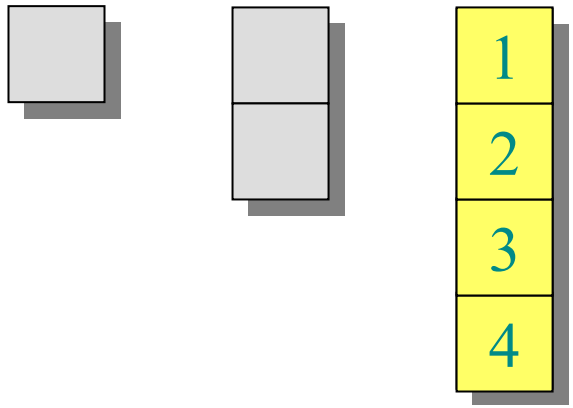
1. INSERT
2. INSERT
3. INSERT




 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Example of a dynamic table

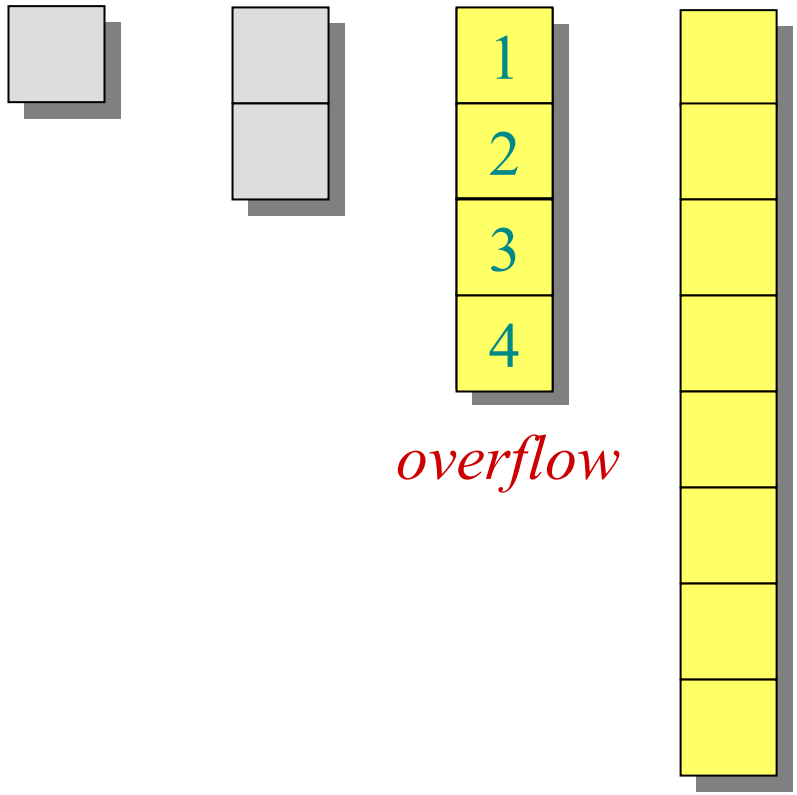
1. INSERT
2. INSERT
3. INSERT
4. INSERT




 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Example of a dynamic table

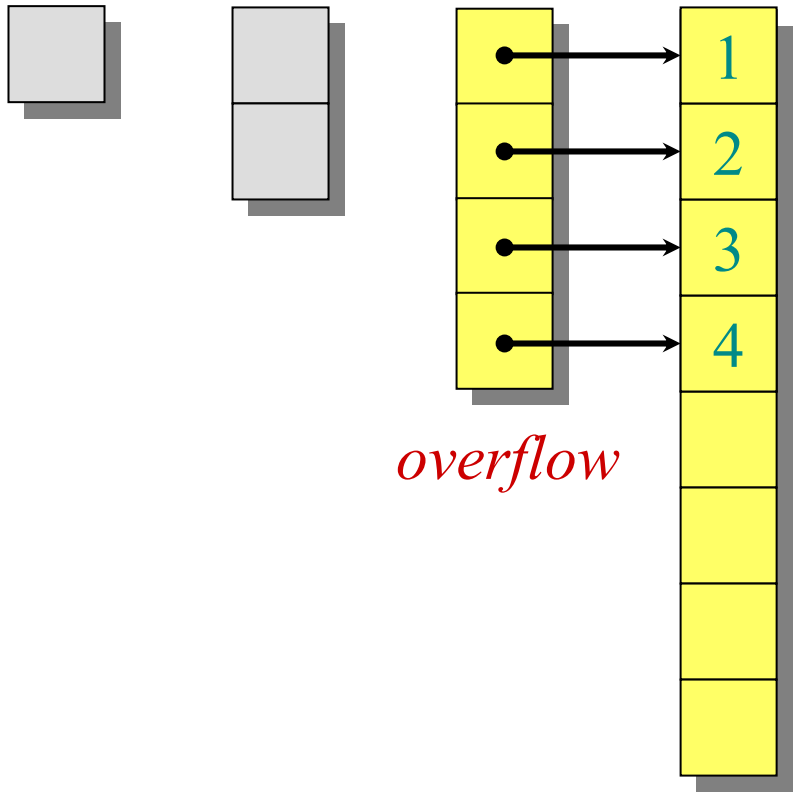
1. INSERT
2. INSERT
3. INSERT
4. INSERT
5. INSERT




 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Example of a dynamic table

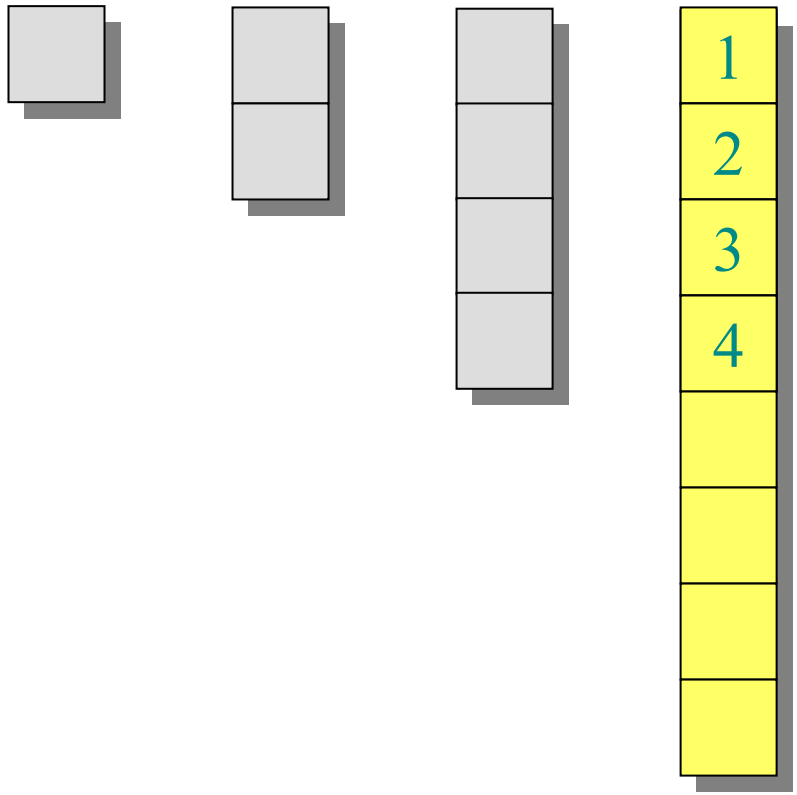
1. INSERT
2. INSERT
3. INSERT
4. INSERT
5. INSERT




 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Example of a dynamic table

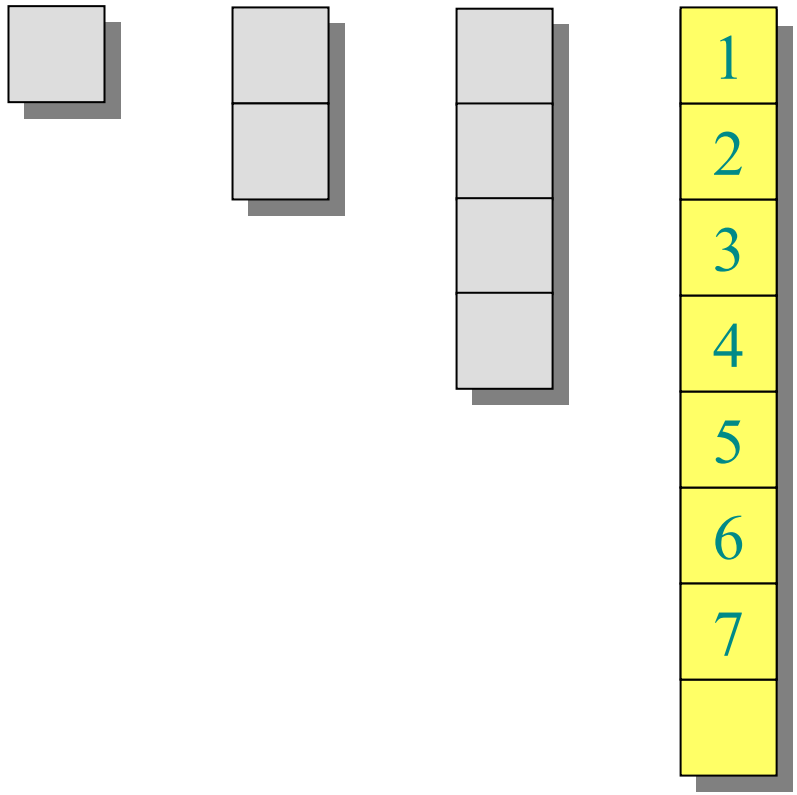
1. INSERT
2. INSERT
3. INSERT
4. INSERT
5. INSERT




 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

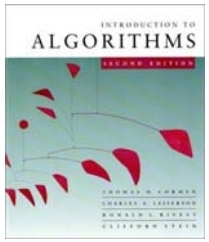
# Example of a dynamic table

1. INSERT
2. INSERT
3. INSERT
4. INSERT
5. INSERT
6. INSERT
7. INSERT





*Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson*




# Worst-case analysis

Consider a sequence of $n$ insertions. The worst-case time to execute one insertion is $\Theta(n)$. Therefore, the worst-case time for $n$ insertions is $n \cdot \Theta(n) = \Theta(n^2)$.

**WRONG!** In fact, the worst-case cost for $n$ insertions is only $\Theta(n) \ll \Theta(n^2)$.

Let's see why.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Tighter analysis

Let $c_i =$ the cost of the $i$th insertion

$$= \begin{cases} i & \text{if } i - 1 \text{ is an exact power of } 2, \\ 1 & \text{otherwise.} \end{cases}$$

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $size_i$ | 1 | 2 | 4 | 4 | 8 | 8 | 8 | 8 | 16 | 16 |
| $c_i$ | 1 | 2 | 3 | 1 | 5 | 1 | 1 | 1 | 9 | 1 |


*Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson*

# Tighter analysis

Let $c_i =$ the cost of the $i$th insertion

$$= \begin{cases} i & \text{if } i - 1 \text{ is an exact power of } 2, \\ 1 & \text{otherwise.} \end{cases}$$

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $size_i$ | 1 | 2 | 4 | 4 | 8 | 8 | 8 | 8 | 16 | 16 |
| $c_i$ | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| | | 1 | 2 | | 4 | | | | 8 | |


*Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson*

# Tighter analysis (continued)

$$\text{Cost of } n \text{ insertions} = \sum_{i=1}^{n} c_i$$

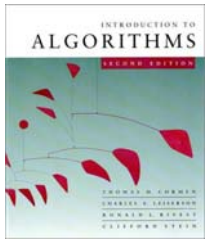
$$\le n + \sum_{j=0}^{\lfloor \lg(n-1) \rfloor} 2^j$$

$$\le 3n$$

$$= \Theta(n).$$

Thus, the average cost of each dynamic-table operation is $\Theta(n)/n = \Theta(1)$.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 



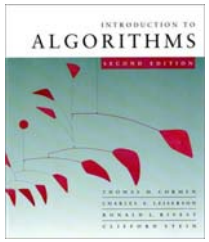

# Amortized analysis

An ***amortized analysis*** is any strategy for analyzing a sequence of operations to show that the average cost per operation is small, even though a single operation within the sequence might be expensive.

Even though we're taking averages, however, probability is not involved!

- An amortized analysis guarantees the average performance of each operation in the *worst case*.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 




# Types of amortized analyses

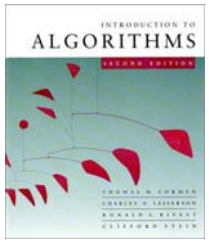
Three common amortization arguments:

- the ***aggregate*** method,
- the ***accounting*** method,
- the ***potential*** method.

We've just seen an aggregate analysis.

The aggregate method, though simple, lacks the precision of the other two methods. In particular, the accounting and potential methods allow a specific ***amortized cost*** to be allocated to each operation.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 




# Accounting method

- Charge $i$th operation a fictitious ***amortized cost*** $\hat{c}_i$, where $1 pays for 1 unit of work (*i.e.*, time).
- This fee is consumed to perform the operation.
- Any amount not immediately consumed is stored in the ***bank*** for use by subsequent operations.
- The bank balance must not go negative! We must ensure that

$$\sum_{i=1}^{n} c_i \le \sum_{i=1}^{n} \hat{c}_i$$

for all $n$.

- Thus, the total amortized costs provide an upper bound on the total true costs.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Accounting analysis of dynamic tables

Charge an amortized cost of $\hat{c}_i = \$3$ for the $i$th insertion.

- $1 pays for the immediate insertion.
- $2 is stored for later table doubling.

When the table doubles, $1 pays to move a recent item, and $1 pays to move an old item.

**Example:**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $0 | $0 | $0 | $0 | $2 | $2 | $2 | $2 |

*overflow*

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | |


*Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson*

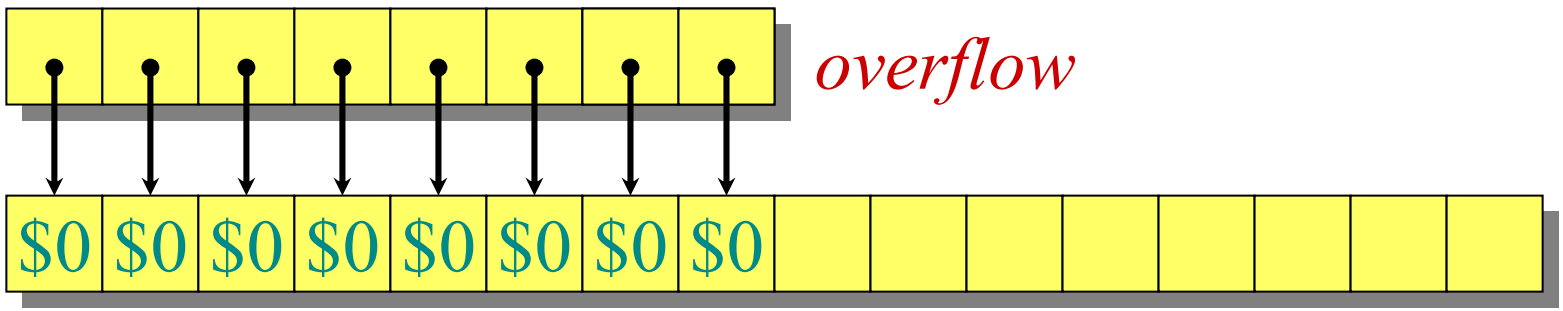
# Accounting analysis of dynamic tables

Charge an amortized cost of $\hat{c}_i = \$3$ for the $i$th insertion.

- $1 pays for the immediate insertion.
- $2 is stored for later table doubling.

When the table doubles, $1 pays to move a recent item, and $1 pays to move an old item.

**Example:**




 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Accounting analysis of dynamic tables

Charge an amortized cost of $\hat{c}_i = \$3$ for the $i$th insertion.

- $\$1$ pays for the immediate insertion.
- $\$2$ is stored for later table doubling.

When the table doubles, $\$1$ pays to move a recent item, and $\$1$ pays to move an old item.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

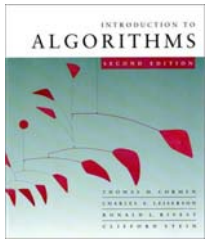
# Accounting analysis (continued)

**Key invariant:** Bank balance never drops below 0. Thus, the sum of the amortized costs provides an upper bound on the sum of the true costs.

| $i$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $size_i$ | 1 | 2 | 4 | 4 | 8 | 8 | 8 | 8 | 16 | 16 |
| $c_i$ | 1 | 2 | 3 | 1 | 5 | 1 | 1 | 1 | 9 | 1 |
| $\hat{c}_i$ | 2* | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| $bank_i$ | 1 | 2 | 2 | 4 | 2 | 4 | 6 | 8 | 2 | 4 |

*Okay, so I lied. The first operation costs only \$2, not \$3.


*Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson*



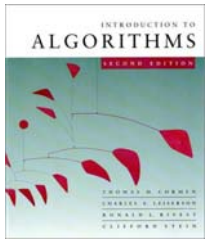

# Potential method

**IDEA:** View the bank account as the potential energy (*à la* physics) of the dynamic set.

**Framework:**

- Start with an initial data structure $D_0$.
- Operation $i$ transforms $D_{i-1}$ to $D_i$.
- The cost of operation $i$ is $c_i$.
- Define a ***potential function*** $\Phi : \{D_i\} \to \mathbb{R}$, such that $\Phi(D_0) = 0$ and $\Phi(D_i) \geq 0$ for all $i$.
- The ***amortized cost*** $\hat{c}_i$ with respect to $\Phi$ is defined to be $\hat{c}_i = c_i + \Phi(D_i) - \Phi(D_{i-1})$.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 




# Understanding potentials

$$\hat{c}_i = c_i + \underbrace{\Phi(D_i) - \Phi(D_{i-1})}_{\textbf{\textit{potential difference}}\ \Delta\Phi_i}$$

- If $\Delta\Phi_i > 0$, then $\hat{c}_i > c_i$. Operation $i$ stores work in the data structure for later use.
- If $\Delta\Phi_i < 0$, then $\hat{c}_i < c_i$. The data structure delivers up stored work to help pay for operation $i$.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# The amortized costs bound the true costs

The total amortized cost of $n$ operations is

$$\sum_{i=1}^{n} \hat{c}_i = \sum_{i=1}^{n} \left( c_i + \Phi(D_i) - \Phi(D_{i-1}) \right)$$

Summing both sides.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# The amortized costs bound the true costs

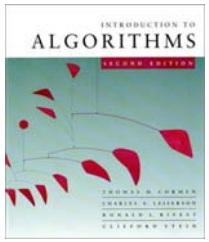
The total amortized cost of $n$ operations is

$$\sum_{i=1}^{n} \hat{c}_i = \sum_{i=1}^{n} \left( c_i + \Phi(D_i) - \Phi(D_{i-1}) \right)$$

$$= \sum_{i=1}^{n} c_i + \Phi(D_n) - \Phi(D_0)$$

The series telescopes.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 




# The amortized costs bound the true costs

The total amortized cost of $n$ operations is

$$\sum_{i=1}^{n} \hat{c}_i = \sum_{i=1}^{n} \left(c_i + \Phi(D_i) - \Phi(D_{i-1})\right)$$

$$= \sum_{i=1}^{n} c_i + \Phi(D_n) - \Phi(D_0)$$

$$\geq \sum_{i=1}^{n} c_i \qquad \text{since } \Phi(D_n) \geq 0 \text{ and } \Phi(D_0) = 0.$$

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

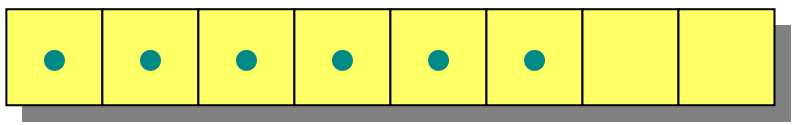
# Potential analysis of table doubling

Define the potential of the table after the ith insertion by $\Phi(D_i) = 2i - 2^{\lceil \lg i \rceil}$. (Assume that $2^{\lceil \lg 0 \rceil} = 0$.)

**Note:**

- $\Phi(D_0) = 0$,
- $\Phi(D_i) \geq 0$ for all $i$.

**Example:**



$\Phi = 2 \cdot 6 - 2^3 = 4$


*Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson*

# Calculation of amortized costs

The amortized cost of the $i$th insertion is

$$\hat{c}_i = c_i + \Phi(D_i) - \Phi(D_{i-1})$$

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Calculation of amortized costs

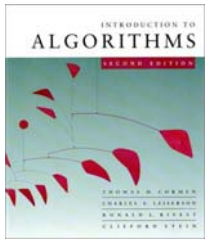
The amortized cost of the $i$th insertion is

$$\hat{c}_i = c_i + \Phi(D_i) - \Phi(D_{i-1})$$

$$= \left\{ \begin{array}{ll} i & \text{if } i-1 \text{ is an exact power of } 2, \\ 1 & \text{otherwise;} \end{array} \right\}$$

$$+ \left(2i - 2^{\lceil \lg i \rceil}\right) - \left(2(i-1) - 2^{\lceil \lg (i-1) \rceil}\right)$$


*Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson*




# Calculation of amortized costs

The amortized cost of the $i$th insertion is

$$\hat{c}_i = c_i + \Phi(D_i) - \Phi(D_{i-1})$$

$$= \left\{ \begin{array}{ll} i & \text{if } i-1 \text{ is an exact power of } 2, \\ 1 & \text{otherwise;} \end{array} \right\}$$

$$+ \left(2i - 2^{\lceil \lg i \rceil}\right) - \left(2(i-1) - 2^{\lceil \lg (i-1) \rceil}\right)$$

$$= \left\{ \begin{array}{ll} i & \text{if } i-1 \text{ is an exact power of } 2, \\ 1 & \text{otherwise;} \end{array} \right\}$$

$$+ 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil}.$$

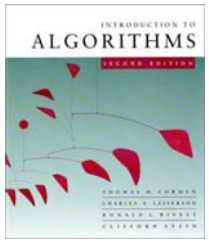
 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Calculation

**Case 1:** $i - 1$ is an exact power of $2$.

$$\hat{c}_i = i + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil}$$


*Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson*

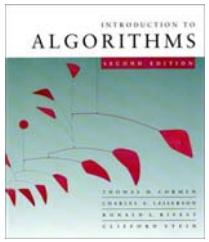



# Calculation

**Case 1:** $i - 1$ is an exact power of $2$.

$$\begin{aligned}\hat{c}_i &= i + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil} \\ &= i + 2 - 2(i-1) + (i-1)\end{aligned}$$

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

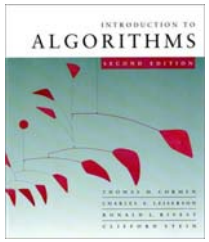



# Calculation

**Case 1:** $i - 1$ is an exact power of $2$.

$$\begin{aligned}\hat{c}_i &= i + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil} \\ &= i + 2 - 2(i-1) + (i-1) \\ &= i + 2 - 2i + 2 + i - 1\end{aligned}$$

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

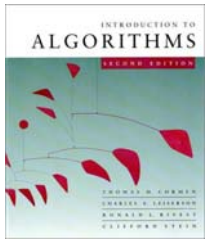



# Calculation

**Case 1:** $i - 1$ is an exact power of $2$.

$$\begin{aligned}\hat{c}_i &= i + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil} \\ &= i + 2 - 2(i-1) + (i-1) \\ &= i + 2 - 2i + 2 + i - 1 \\ &= 3\end{aligned}$$

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 




# Calculation

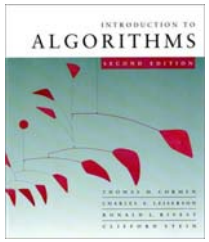
**Case 1:** $i - 1$ is an exact power of $2$.

$$\begin{aligned}\hat{c}_i &= i + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil} \\ &= i + 2 - 2(i-1) + (i-1) \\ &= i + 2 - 2i + 2 + i - 1 \\ &= 3\end{aligned}$$

**Case 2:** $i - 1$ is *not* an exact power of $2$.

$$\hat{c}_i = 1 + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil}$$

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 




# Calculation

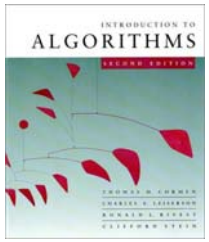
**Case 1:** $i - 1$ is an exact power of $2$.

$$\begin{aligned}\hat{c}_i &= i + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil} \\ &= i + 2 - 2(i-1) + (i-1) \\ &= i + 2 - 2i + 2 + i - 1 \\ &= 3\end{aligned}$$

**Case 2:** $i - 1$ is *not* an exact power of $2$.

$$\begin{aligned}\hat{c}_i &= 1 + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil} \\ &= 3 \qquad\qquad (\text{since } 2^{\lceil \lg i \rceil} = 2^{\lceil \lg (i-1) \rceil})\end{aligned}$$

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 




# Calculation

**Case 1:** $i - 1$ is an exact power of $2$.

$$\begin{aligned}\hat{c}_i &= i + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil} \\ &= i + 2 - 2(i-1) + (i-1) \\ &= i + 2 - 2i + 2 + i - 1 \\ &= 3\end{aligned}$$

**Case 2:** $i - 1$ is *not* an exact power of $2$.

$$\begin{aligned}\hat{c}_i &= 1 + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil} \\ &= 3\end{aligned}$$

Therefore, $n$ insertions cost $\Theta(n)$ in the worst case.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 

# Calculation

**Case 1:** $i - 1$ is an exact power of $2$.

$$\begin{aligned}\hat{c}_i &= i + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil} \\ &= i + 2 - 2(i-1) + (i-1) \\ &= i + 2 - 2i + 2 + i - 1 \\ &= 3\end{aligned}$$

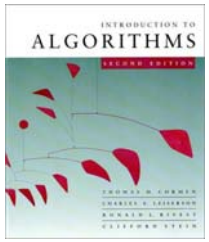
**Case 2:** $i - 1$ is *not* an exact power of $2$.

$$\begin{aligned}\hat{c}_i &= 1 + 2 - 2^{\lceil \lg i \rceil} + 2^{\lceil \lg (i-1) \rceil} \\ &= 3\end{aligned}$$

Therefore, $n$ insertions cost $\Theta(n)$ in the worst case.

**Exercise:** Fix the bug in this analysis to show that the amortized cost of the first insertion is only $2$.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 




# Conclusions

- Amortized costs can provide a clean abstraction of data-structure performance.
- Any of the analysis methods can be used when an amortized analysis is called for, but each method has some situations where it is arguably the simplest or most precise.
- Different schemes may work for assigning amortized costs in the accounting method, or potentials in the potential method, sometimes yielding radically different bounds.

 *Copyright © 2001-5 by Erik D. Demaine and Charles E. Leiserson* 